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SoundRate: Distributed Acoustic Sensing for Autonomous Flow Allocation via Unsupervised Learning

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Abstract

An unsupervised methodology is introduced for quantitative zonal rate allocation in wellbores equipped with temporary or permanent fiber optic cables for acoustic sensing. This approach aims to mitigate key limitations of conventional data-driven methods, which rely heavily on labeled training data and prior knowledge and often struggle to accommodate temporal and spatial variability in acoustic signatures originating from perforations or flow control devices distributed across the lower completion. The proposed framework employs an adaptive, unsupervised learning model capable of autonomous zonal flow allocation using distributed acoustic sensing data. At each rate change event, the algorithm leverages differential acoustic responses along the fiber, identifying how zone-specific signatures evolve relative to total flow variations. This enables the development of unique flow-acoustic relationships for individual zones without prior assumptions. Furthermore, the self-learning model continuously updates zonal contributions, adapting to changing downhole environments and completion characteristics. The model is applicable across all completion types—including open hole scenarios—provided acoustic data is available. Validation using synthetic wells demonstrates successful allocation of flow rates across multiple zones, aligning with expected injection and production profiles. This technique offers potential value for reservoir management, production optimization, and well integrity monitoring by enabling continuous, quantitative zonal flow allocation. The novelty of the method lies in its unsupervised learning framework, which leverages acoustic data during rate change events to dynamically calibrate the relationship between flow-induced noise and zonal flow rates—surpassing the limitations of empirical and supervised models. Beyond allocation, the learned acoustic-flow relationships offer diagnostic value, enabling insights into completion performance and zonal behavior over time.

Introduction

Effective reservoir management—including well stimulation planning and production optimization—depends on understanding the dynamic contributions of individual zones to wellbore flow. Accurate allocation of injected or produced fluid rates, especially during interventions or enhanced oil recovery (EOR) operations, is essential for verifying completion efficiency and identifying underperforming zones. It also plays a critical role in detecting unwanted water or gas breakthrough and improving hydrocarbon recovery.

However, in complex wells with multiple zones, extended contributing sections, and advanced flow control devices, obtaining a reliable and continuous zonal flow profile remains a persistent challenge. Over the years, a range of technologies has been employed—each with distinct capabilities and limitations. From traditional intervention-based tools to downhole gauges and, more recently, fiber optic sensing, the pursuit of reliable flow profiling continues to evolve.

Traditional flow profiling methods, such as Production Logging Tools (PLTs), have long provided valuable and direct insights into zonal contributions to flow. A typical PLT string incorporates multiple sensors—including spinners, pressure and temperature gauges, and fluid holdup detectors—deployed via wireline or coiled tubing to collect measurements at various depths (Chowdhury *et al.* 2019). By analyzing this data, engineers can construct detailed flow profiles: spinners measure fluid velocity, which, when integrated with holdup and pressure readings, enables estimation of zonal flow rates. While PLTs offer high-resolution, depth-specific data, their application comes with notable constraints. PLT operations are resource-intensive, requiring specialized crews and equipment, which can lead to deferred production and elevated operational costs. Additionally, each PLT run captures only a momentary snapshot of well conditions, limiting its effectiveness in tracking real-time response to operational changes, transient flow behaviors or long-term performance trends. Gathering additional data necessitates repeat interventions, compounding both cost and risk. In wells that are highly deviated, contain mechanical restrictions, or are situated in offshore environments, PLT deployment may be technically challenging or even unfeasible. These practical limitations have driven the industry toward complementary and alternative solutions that enable continuous monitoring, reduce operational footprint, and improve economic efficiency—while preserving the diagnostic value PLTs continue to offer.

Recent advances in fiber optic monitoring—particularly Distributed Acoustic Sensing (DAS)—have opened new possibilities for continuous, real-time downhole surveillance. The optical fiber functions as a continuous sensor along the entire wellbore. DAS operates based on Rayleigh backscattering: a laser interrogator at the surface sends pulses of light down the fiber, which are scattered by natural imperfections in the glass. When an acoustic field—such as fluid flow or vibration—occurs near the fiber, it induces minute strain that alters the phase of the backscattered light. By analyzing the timing and phase of the returning signal, the interrogator can pinpoint and characterize acoustic events in real time with high spatial and temporal resolution. The noise generated by fluid flowing through completions—such as perforations, ICDs, or valves—is inherently linked to the flow itself. This connection has inspired extensive research into using DAS as a proxy for PLT-based flow monitoring (Van der Horst *et al.* 2014, Liu *et al.* 2024).

While DAS has shown promise for qualitative diagnostics, its use in quantitative flow allocation has been constrained by several critical limitations. The available methods can be summarized into data-driven and physics-based groups. Physics-based methods such as Speed of Sound (SoS) or Doppler shift tracking use DAS to quantify single- and multiphase flow rates by analyzing acoustic wave propagation to infer fluid velocity (Xiao *et al.* 2015). These methods, however, rely on the ability to detect high-fidelity, broadband flow noise, which is often difficult to isolate from complex, non-coherent background noise. This challenge of measuring SoS with a very high accuracy is particularly pronounced in unconventional wells, where flow-induced acoustic signals tend to be weak or intermittent. Furthermore, SoS measurements are typically more accurate when long-array processing techniques, such as frequency–wavenumber (FK) analysis, are employed. While these methods enhance measurement confidence, they inherently reduce spatial resolution—limiting the ability to resolve fine-scale zonal contributions. Another robust physics-based approach involves tracking the movement of low-frequency features, commonly referred to as plume tracking. This technique monitors transient thermal or acoustic plumes generated during changes in production or injection, enabling direct mapping of flow velocity across different sections of the well. While promising, plume tracking—like PLTs—provides only a snapshot of well behavior. The flow dynamics captured during such

transient events may differ significantly from those observed under steady-state production or injection conditions, limiting its effectiveness for continuous monitoring.

Compared to physics-based methods, data-driven approaches require fewer computational resources, enable real-time solutions, offer superior spatial resolution, and exhibit lower sensitivity to noise. Within this category, supervised or empirical techniques aim to learn the relationship between acoustic noise and flow rate directly from data. These models rely on a ground truth dataset for training, where ground truth refers to known zonal flow rates paired with corresponding DAS acoustic measurements. The reliance on labeled downhole data, however, presents a practical barrier where such data are difficult to obtain. Acoustic signatures along the lower completion vary due to factors such as completion design (e.g., perforations, valves, screens), local coupling effects, and operational parameters like pressure and flow regime. These signatures also evolve over time due to erosion, scaling, or mechanical wear, introducing data drift. To maintain model accuracy, labeled datasets must be updated frequently, which requires repeated PLT runs solely for DAS model calibration. This undermines the very goal of replacing PLT interventions. Without labeled training data, supervised models cannot be effectively developed, extrapolated, or validated for a specific well.

Given these challenges, there is a clear need for quantitative flow allocation methods that avoid static physical assumptions, eliminate the dependency on frequent downhole data acquisition for calibration, generalize effectively without overfitting, and remain robust under low signal-to-noise conditions. Advancing such approaches would enable continuous, non-intrusive, and cost-efficient flow monitoring—unlocking the full potential of DAS as a practical alternative to conventional PLT-based surveillance.

This work introduces SoundRate—an adaptive, unsupervised learning framework for quantitative fluid rate allocation using DAS data. Leveraging a data-driven noise logging strategy, the method capitalizes on rate change events in total well flow to infer zone-specific relationships between flow and acoustic response. By systematically analyzing differential acoustic signals along the fiber during each rate transition, SoundRate identifies unique acoustic signatures for each completion zone and constructs individualized flow–acoustic performance curves. Crucially, this is achieved without reliance on pre-calibrated models or labeled datasets. The following section outlines the methodology and demonstrates its performance and generalizability through synthetic case studies.

Methodology

McKinley *et al.* (1973, 1977) conducted experimental studies on noise logging of single- and two-phase fluid flow behind casing and established a model to correlate the noise amplitude with the flow rate, applicable to both flow behind the casing and flow through perforations. McKinley's model to correlate noise and single-phase gas or liquid volumetric flow rate reads,

$$N = C\rho\frac{q^3}{A^2} \quad (1)$$

where N is the peak-to-peak noise level above 600 Hz in millivolts, ρ represents fluid density, A is the cross-sectional area open to flow, and, q and C are the volumetric flow rate and the correlation coefficient equivalent to a drag coefficient, respectively. The model suggests a cubic relationship between noise level and gas or liquid flow rate and a linear relationship between density and noise level. It should be noted that viscosity effect manifests itself in the correlation constant. This cubic correlation has been widely employed and validated in other experimental and numerical studies utilizing acoustic measurements to quantify fluid flow through fractures, perforations and AICDs (Frieauf *et al.* 2019, Hamanaka *et al.* 2024, Pakhotina *et al.* 2017, Chen *et al.* 2015, Moradi *et al.* 2024).

Although the cubic relationship between noise and rate has been validated for multiple applications, the parameters C , A and ρ shape the acoustic signature and are highly sensitive to completion design, fluid characteristics, and local operating conditions. These parameters are dynamic and may evolve over time due

to perforation erosion, scaling, or changes in fluid composition. As previously discussed, a key limitation in applying empirical models like Eq.1 lies in the need for sufficient labeled data from each well section to calibrate the model accurately. This same limitation extends to supervised machine learning approaches for flow allocation, which similarly rely on high-quality labeled datasets to achieve reliable performance.

In this work, the methodology is designed to derive quantitative zonal flow rates by learning the unique relationship between acoustic energy and flow for each zone, without prior assumptions or labeled data. The core principle is that while the individual zonal contributions are unknown, their sum is known. By observing how zone-specific acoustic signals respond to changes in the total rate, the algorithm can solve for the individual contribution factors. The method is predicated on a simple mass-balance constraint. For a well with N distinct inflow or injection zones, the total liquid flow rate at bottom-hole conditions, $Q(t)$, is the sum of the individual zonal flow rates (q_i):

$$Q(t) = \sum_{i=1}^N q_i(t) \quad (2)$$

The central hypothesis of method is that the flow rate for a given zone is a function of its acoustic pressure, governed by zone-specific coefficients. We model this relationship as a power law, which is consistent with the physics of flow-generated sound but avoids assuming a universal exponent:

$$q_i(t) = c_i A_i^{2n} P_i(t)^n \quad (3)$$

where c_i is the unknown zonal contribution coefficient that encapsulates the unique geometric and fluidic properties of that specific zone. The exponent n represents the general physical relationship between acoustic pressure and flow rate. While n could also be a variable, for robustness and to avoid overfitting, it is often set to a physically plausible constant based on Eq.1. The coefficient c_i is treated as a unique unknown for each zone. Substituting Eq.3 into Eq.2 gives:

$$Q(t) = \sum_{i=1}^N c_i A_i^{2n} P_i(t)^n \quad (4)$$

Eq.4 forms the basis of an inverse problem. We have the measured time-series $Q(t)$ and $P_N(t)$, and our goal is to find the set of unknown coefficients c_i . Solving the system defined by Eq.4 requires two synchronized time-series datasets collected during an operational change in the total flow rate—such as a step-rate test or a production onset. These include: (1) the total bottom-hole injection or production rate, and (2) DAS measurements recorded along the lower completion, encompassing all zones of interest. While DAS inherently provides high-frequency data, the total flow rate must also be captured at a sufficiently high temporal resolution to generate enough data points for robust model training.

The raw DAS data, recorded strain rate along the lower completion, is processed to yield a meaningful metric for flow-induced noise at each zone. This involves filtering and normalization. For each of the N zones, a specific depth interval along the fiber is defined. The acoustic data within this interval is then filtered to a frequency band relevant to flow noise in three space, time and frequency-wavenumber domains to remove extraneous noise from mechanical sources or far-field effects. The zone noise is then normalized with respect to a reference point to further localize the flow-related noise at each completion zone. For each zone, the filtered acoustic data is transformed into a time series of acoustic power. This is typically computed using the RMS or variance of the acoustic amplitude over short time windows (e.g., 30 seconds), providing a measure of the equivalent continuous level of a fluctuating sound that accounts for both amplitude and duration. Once the RMS values are obtained, the Sound Pressure Level (SPL) is calculated in decibels (dB), offering a logarithmic representation of the acoustic intensity. The output of this stage is N time-series vectors, $P_1(t), \dots, P_N(t)$, each representing the acoustic pressure of a specific zone, time-aligned with the $Q(t)$ vector.

In an ideal scenario, rate and acoustic measurements are acquired across multiple time steps with varying flow conditions. This yields a large set of simultaneous equations, forming an overdetermined system that

can be solved with high robustness. The optimal set of coefficients c_i is determined by minimizing the discrepancy between the measured total flow rate and the model-estimated total flow rate. This calibration is framed as a non-linear least-squares optimization problem, where the objective is to minimize the sum of squared errors across all Mtime steps in the dataset:

$$\min_{c_1, \dots, c_N} \sum_{t=1}^M \left[\left(\sum_{i=1}^N c_i A_i^{2n} P_i(t^n) \right) - Q_{\text{total}}(t) \right]^2 \quad (5)$$

The optimization is performed using a non-linear least-squares solver—typically Levenberg–Marquardt or trust-region reflective methods—selected based on convergence behavior and sensitivity characteristics. The goal is to identify a single set of coefficients that best fits the data across the entire step-rate test. To enhance robustness, proprietary procedures are implemented to mitigate convergence toward local minima, thereby improving the reliability and repeatability of the solution. Once the optimal coefficients are established, the calibrated relationship enables direct computation of zonal flow rates at any time step, yielding time-resolved flow estimates across all zones of interest.

The complete computational workflow is summarized in the flowchart in Fig.1. It begins with pre-processing of synchronized rate and acoustic datasets to extract high-quality, time-aligned data points. Signal conditioning—including bandpass filtering, RMS computation, and zone-specific depth interval mapping—is applied to isolate flow-relevant acoustic features. A critical diagnostic then assesses whether the system is well-posed. If not, a simplified model with a shared coefficient is applied to preserve stability. For well-posed systems, the algorithm proceeds by identifying and grouping zones with correlated acoustic behavior before solving the primary inversion problem. To quantify solution confidence, a Jacobian-based identifiability score is computed. This score reflects the rank of the Jacobian matrix, which measures the degree of linear independence among the zonal acoustic response vectors. A low rank indicates high correlation and an ill-conditioned system, while a high rank confirms that the data contains sufficient independent information to resolve a unique set of coefficients. This scoring mechanism serves as a critical quality control gate, preventing convergence toward non-unique or physically implausible solutions and ensuring the stability and diagnostic integrity of the autonomous allocation process.

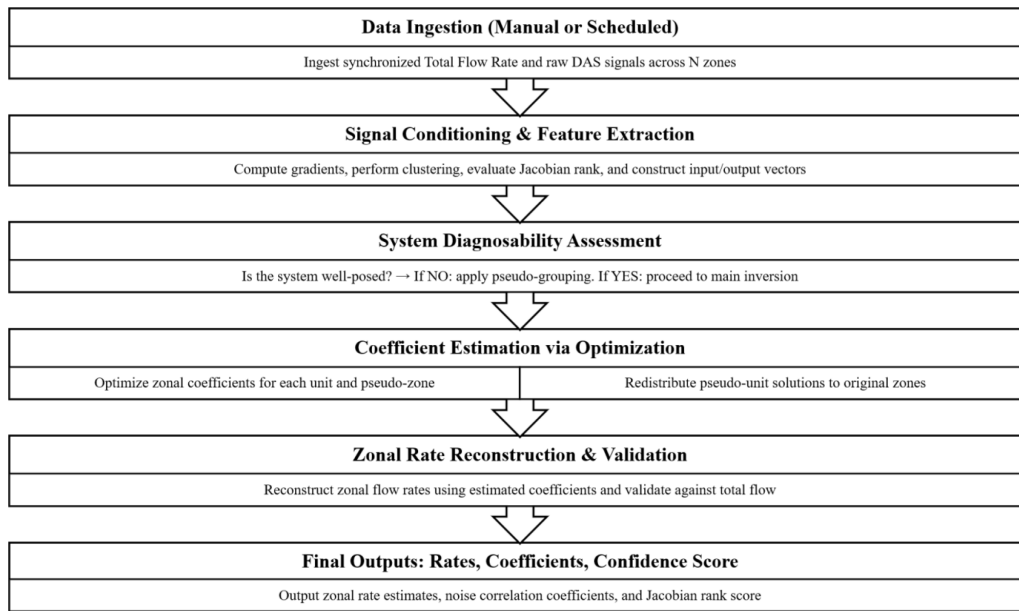


Figure 1—Flowchart of the computational workflow. The diagram outlines the key stages: synchronized data pre-processing, a diagnostic check for system solvability, zone grouping based on acoustic correlation, execution of the primary inversion to allocate zonal flow rates, and a final Jacobian-based identifiability analysis to quantify solution confidence.

Application Criteria

The methodology's ability to resolve individual zonal contributions hinges on the presence of transient flow redistributions, which generate distinct, time-varying acoustic signatures essential for a well-posed inversion. These transients may be induced—via production onsets, step-rate tests, or localized choke adjustments in intelligent completions—or may arise naturally from reservoir dynamics. In the latter case, non-linearities in zonal pressure-flow behavior, driven by near-wellbore effects, mobility contrasts, or rate-dependent relative permeability, can cause evolving flow distributions. The algorithm is designed to capitalize on either type of transient—induced or spontaneous—to continuously refine its understanding of zonal acoustic signatures and improve flow allocation accuracy.

The inversion process relies on two key criteria: (1) a sufficient number of rate change events relative to the number of zones, and (2) distinct acoustic responses across zones during each event. The workflow can be triggered during diagnostic operations or run continuously in real-time, adapting as new rate changes occur. If the incoming data is insufficient to yield a unique solution, the system defers model updates until additional rate steps improve the conditioning. In cases where multiple zones exhibit highly similar acoustic signatures, the algorithm can automatically group them and solve for a single coefficient set per group. In essence, the resolution of the reconstructed rate profile scales with the number and diversity of rate change datapoints—more rate steps yield finer granularity.

As discussed, robust signal conditioning is essential to isolate meaningful acoustic features and ensure the inversion remains sensitive to zonal flow variations. Additionally, the effectiveness of transient detection depends on the temporal resolution of the rate measurements. Rapid rate changes must be captured with sufficient fidelity to preserve the acoustic signature evolution across zones. This requires careful synchronization between bottomhole acoustic sensing and surface rate measurements, as delays introduced by wellbore volume, fluid compressibility, and sensor response can distort the alignment between rate steps and acoustic features. To mitigate these effects, the averaging interval or calculation window used in the inversion can be dynamically adjusted to accommodate known delays, stabilize signal interpretation, and maintain temporal coherence across data streams.

Results and Discussion

To validate its performance, the proposed methodology was applied to four synthetic analytical and numerical case studies. These cases model wells with multiple contributing zones, with each study representing a unique performance scenario. A key constraint of the problem is that while the total injection or production rate is known—measured at the surface and converted to bottom-hole conditions—the individual contribution of each zone is unknown and needs to be estimated. The subsequent sections present the results from these case studies, highlighting the algorithm's effectiveness in diagnosing specific performance scenarios and accurately allocating the zonal injection rates.

Case 1: Step-Rate Test in a Production Well

This case simulates a five-step rate test in a single-phase vertical production well with two contributing zones, each with a thickness of 30 m, designed to validate the algorithm's core functionality under ideal pseudo-steady-state conditions. Zone productivities are predefined, and the total well production is increased incrementally in five steps, each lasting 10 minutes. In the forward model, zonal flow rates are first computed analytically based on the prescribed zone productivities and total well rate. Using zone-specific acoustic coefficients, we then simulate the corresponding noise contributions from each zone, generating synthetic noise data tied to the total flow sampled every 20 seconds. The inverse model takes the total flow rate and the composite noise signal as inputs and applies the presented algorithm to estimate the correlation coefficients for each zone. As shown in Fig. 2, the reconstructed rates match the prescribed values with high fidelity, and the estimated noise contribution coefficients closely align with their true predefined values. The

Jacobian rank is calculated to be 83%, confirming that the inverse problem is well-posed and the solution is stable.

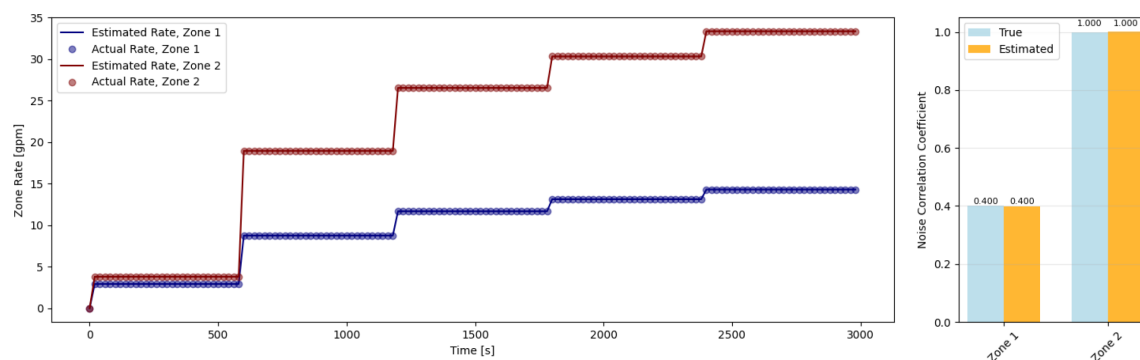


Figure 2—Validation for Case 1 (Step-Rate Test). Comparison of estimated and actual zonal rates (left) and corresponding normalized noise correlation coefficients (right). The reconstructed rates show excellent agreement with the prescribed values, and the estimated coefficients closely match their true counterparts, validating the core inversion workflow under ideal, controlled conditions.

Fig.3 presents a sample of actual DAS data acquired during a rate change event, closely resembling the scenario analyzed in this case study. The top plot illustrates the energy level shift across 50 DAS channels triggered by the rate change across the pipeline (indicated by the dashed line)—an event that DAS detects almost instantaneously. The bottom plot displays a zoomed-in view of the raw strain rate data captured by the system, where sound waves propagating in opposite directions are clearly distinguishable.

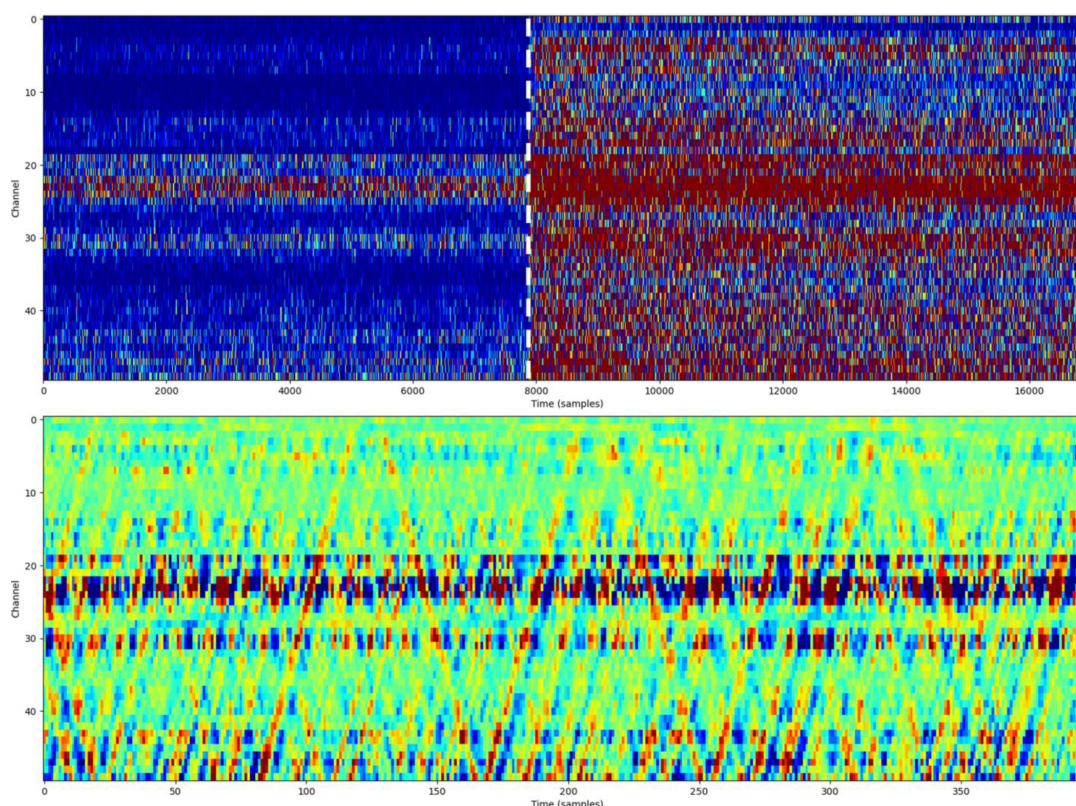


Figure 3—Sample DAS response to a rate change event. (Top) Energy level shift across 50 DAS channels, with the rate change marked by the dashed line. (Bottom) Zoomed-in raw strain rate data showing bidirectional acoustic wave propagation.

Case 2: Continuous Rate Ramp-Up in a Horizontal Well

This case study simulates production ramp-up in a horizontal well with five distinct producing zones, each 150 m in length. As illustrated in Fig.4, flow from each zone is managed by a flow control device. To create a realistic forward model, the inflow performance of each zone was defined by coupling the reservoir model with an analytical model for the valve. Each valve introduces a rate-dependent pressure drop, which, combined with unique zonal reservoir behavior, results in a distinct inflow performance relationship for each zone. The reservoir and bottom-hole pressures are 7000 psi and 4500 psi, respectively. Other formation properties include a porosity of 15%, thickness of 150 ft, fluid viscosity of 10 cP, and a formation volume factor of 1.2. As the well is opened and the total rate ramps up, these differing IPRs cause each zone to contribute uniquely to the total flow, creating the transient conditions necessary for the inversion. Furthermore, each zone is assigned a unique acoustic signature to simulate the distinct noise generated by its specific valve and completion geometry.

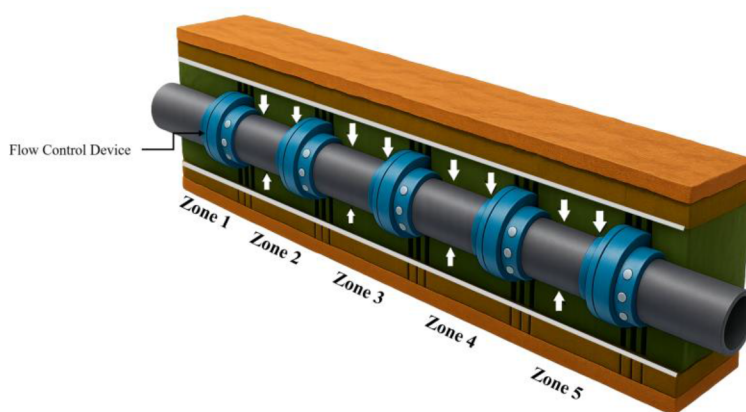


Figure 4—Schematic of the synthetic horizontal well for Case 2, featuring five 150 m producing zones, each equipped with a unique flow control device to create distinct inflow performance relationships.

As shown in Fig.5, applying the presented inversion workflow, the noise correlation coefficients are accurately estimated, enabling precise reconstruction of the zonal flow rates. The methodology demonstrates strong robustness under transient conditions. The Jacobian rank in this case is calculated to be 93%, placing it within the excellent identifiability range.

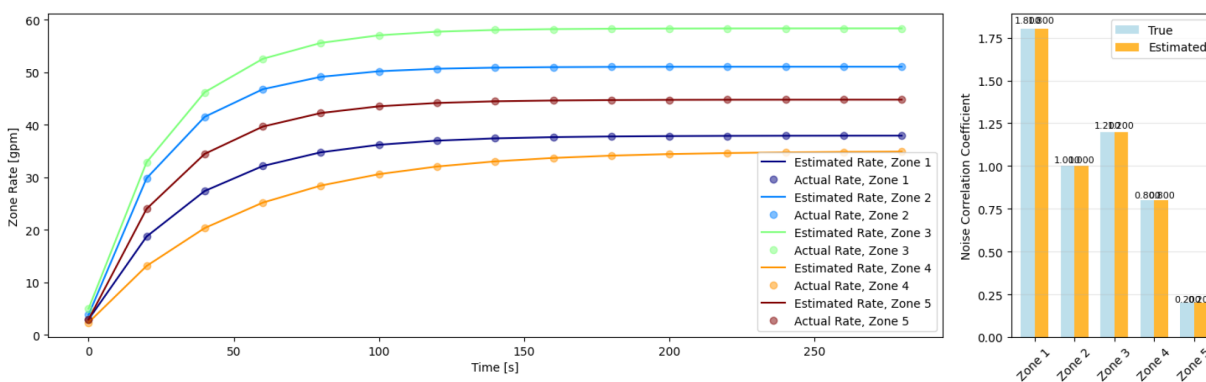


Figure 5—Validation for Case 2 (Production Ramp-Up). The algorithm accurately allocates flow during the transient ramp-up, achieving a high Jacobian rank (93%) due to the distinct hydraulic behavior of each zone. The estimated rates (left) and normalized coefficients (right) closely match the true values.

It should be noted that, in Fig.5, the near-perfect matches and high Jacobian rank are attributed to the distinct ramp-up behavior of each producing zone. However, for instance, if the reservoir properties of Zones

2 and 3 are modified to produce identical flow ramp-up behavior during rate change events—while retaining different acoustic signatures—the algorithm defines a pseudo-zone and estimates an average rate between the two zones (Fig.6). This outcome reflects a key aspect of the workflow: when hydraulic behavior is indistinguishable during rate-change events, the model converges toward a composite solution that preserves mass balance while honoring acoustic differentiation between the zone behaving similarly.

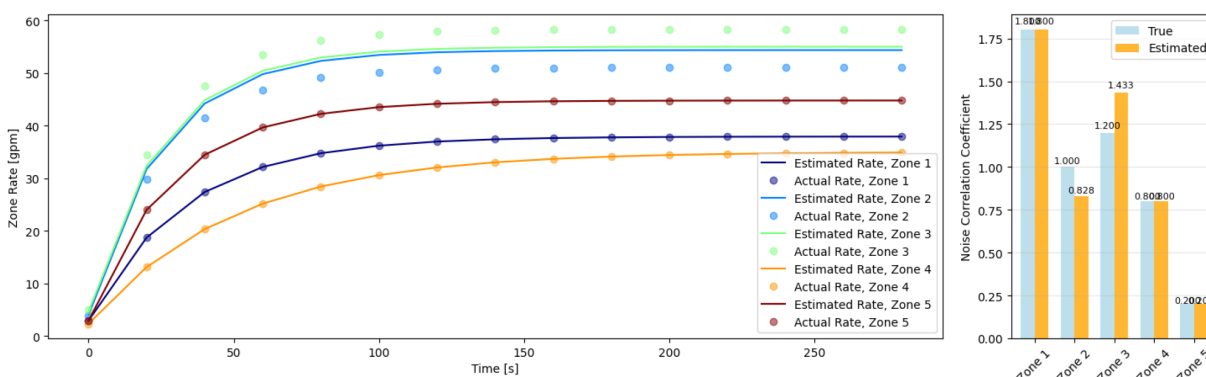


Figure 6—Demonstration of pseudo-zone allocation for a modified Case 2. When Zones 2 and 3 are set to have identical hydraulic behavior but distinct acoustic signatures, the algorithm identifies them as a single "pseudo-zone" and allocates an average rate, while preserving the overall mass balance.

Case 3: Well Stimulation

This case study was designed to test the methodology's ability to adapt to dynamically changing downhole conditions, simulating a well intervention such as a stimulation treatment. In this scenario, fluid is injected at a constant surface pressure into a three-zone pay interval of a vertical well. The key feature of this case is that the permeability of each zone evolves throughout the injection, mimicking the effect of the treatment. This change in reservoir properties creates a continuous, organic variation in the individual zonal injection rates, providing an ideal scenario to validate the workflow's autonomous tuning capability.

As depicted in Fig.7, each zone responds differently to the treatment, resulting in distinct, continuously evolving rate profiles over the two-hour injection period. The rich transient data generated by these dynamic changes allowed the algorithm to accurately estimate the prescribed noise correlation coefficients and reconstruct the zonal flow profiles. This case demonstrates the algorithm's robustness in a scenario where rate changes are not deliberately induced but are a natural consequence of changing well conditions, highlighting its potential for real-time monitoring during complex well operations.

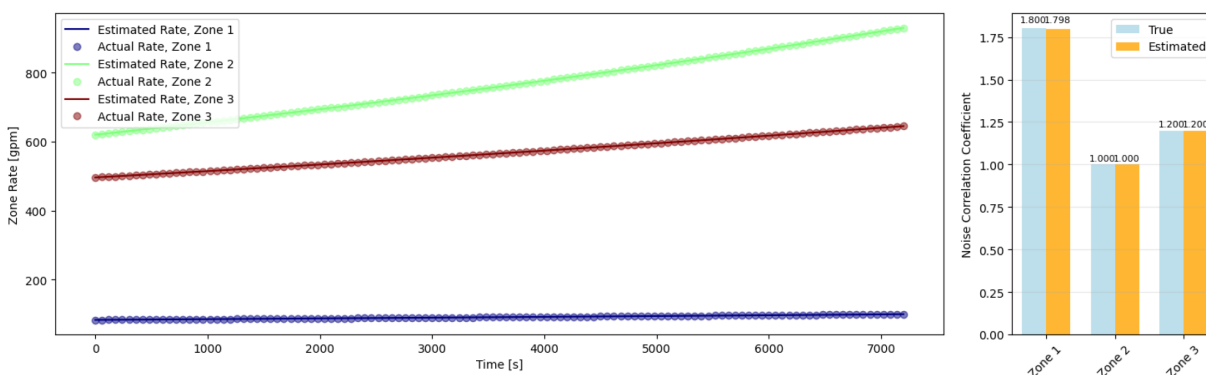


Figure 7—Validation for Case 3 (Simulated Well Stimulation). The algorithm successfully tracks the continuously evolving zonal injection rates driven by organic changes in permeability, demonstrating its adaptability to dynamic downhole conditions without requiring discrete rate steps.

Case 4: Production Onset in a Vertical Well

This case study numerically models pressure drawdown and zonal production behavior in a vertically perforated well intersecting a three-layer reservoir. The simulation spans 10 days and captures dynamic pressure evolution using a radial inflow model. The reservoir comprises three layers with permeability values of 90 mD, 60 mD, and 10 mD respectively (Fig.8). Pressure is initialized uniformly at 7000 psi, with a bottom-hole pressure of 4500 psi driving flow. The diffusivity map, derived from permeability, governs pressure propagation across the domain. The wellbore, centrally located and perforating all layers, enables direct comparison of zonal contributions and transient drawdown effects. To evaluate the algorithm's robustness under realistic sensing conditions, synthetic noise is added to both the DAS signal and the zonal rate data prior to inversion.

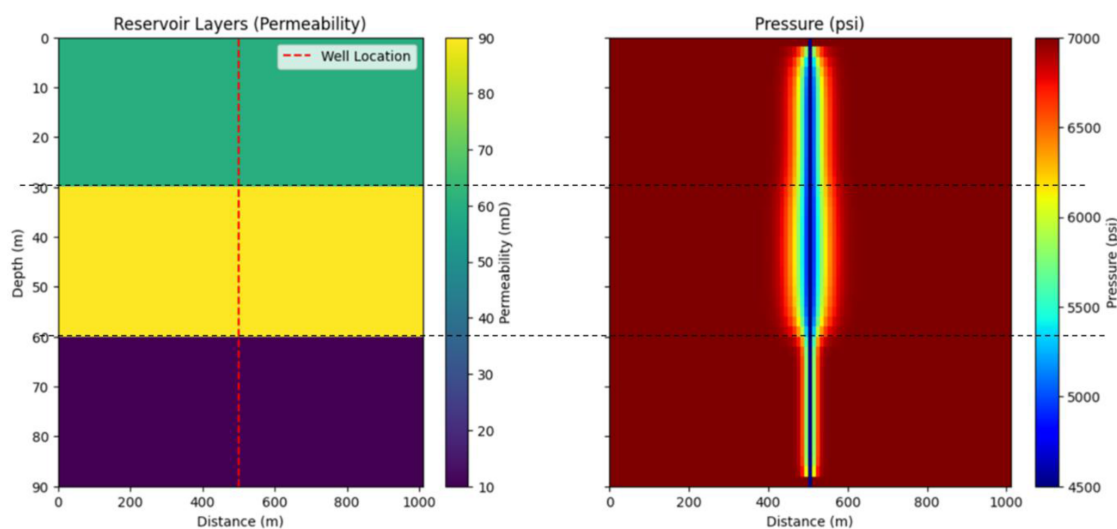


Figure 8—Simulated radial reservoir model showing permeability distribution (left) and pressure drawdown profile (right) for Case 4. The pressure map reflects the imposed bottom-hole pressure, initiating radial inflow and transient propagation across the stratified domain.

Fig.9 illustrates the evolving zonal production behavior in response to natural drawdown dynamics across the three-layer reservoir. Unlike scenarios with externally induced rate perturbations, the observed flow profiles emerge organically from pressure gradients and layer-specific permeability contrasts. Each zone exhibits a distinct transient signature, shaped by its diffusivity and connectivity to the wellbore. These variations enable the algorithm to resolve zonal contributions with high fidelity, even under continuous, non-stepped flow conditions. The successful reconstruction of layer-specific rates and pressure fields underscores the algorithm's adaptability to realistic reservoir heterogeneity and its suitability for real-time diagnostics in vertically perforated wells.

It is important to note that the core algorithm is designed to allocate the total liquid rate for each zone. For multiphase applications, such as oil-water flow, an additional step is required to determine the volume fraction of each phase. The total zonal liquid rate from the method can be combined with an independent measurement of the zonal water cut to deconvolve the individual phase rates. This water cut estimate can be derived from complementary analysis of the same DAS data—for instance, by calculating the SoS—or from material balance calculations (Moradi 2025). This integrated workflow enables the construction of a complete quantitative profile of both oil and water flow for each zone.

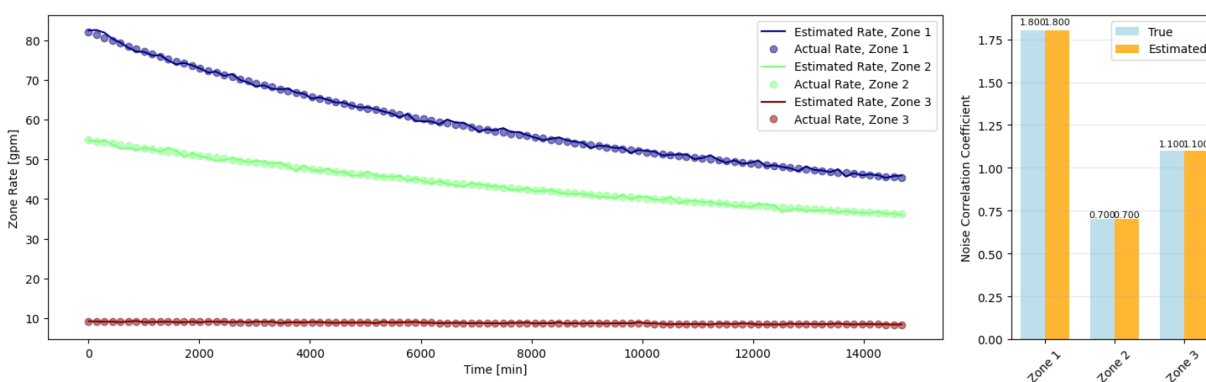


Figure 9—Validation for Case 4 (Natural Drawdown). The algorithm accurately resolves the zonal production profiles using only the organic transients generated by reservoir pressure depletion, highlighting its capability for long-term, non-interventive monitoring

Conclusion

This paper introduced SoundRate, an adaptive, unsupervised learning framework designed to overcome the critical limitations of conventional DAS-based flow allocation methods. By capitalizing on the differential acoustic responses during rate change events—whether induced or naturally occurring—the methodology autonomously calibrates unique flow-acoustic relationships for each zone without requiring labeled data from PLT interventions. The robustness and accuracy of the approach were demonstrated through a series of diverse synthetic case studies. The algorithm successfully allocated rates in scenarios ranging from simple step-rate tests to complex dynamic events, including production ramp-ups with flow control devices, simulated well stimulations with evolving permeability, and production driven by natural reservoir drawdown. The primary contribution of this work is the elimination of the dependency on costly and operationally intensive calibration runs, presenting a clear pathway toward continuous, cost-effective, and autonomous zonal surveillance. Beyond simple allocation, the methodology transforms DAS into a powerful diagnostic tool. By monitoring the learned flow-acoustic coefficients over time, operators can gain insights into the health of the completion, detecting potential integrity issues such as erosion, plugging, or scaling. This capability elevates the technique from a measurement tool to a proactive system for data-driven well integrity management, unlocking the full potential of permanent fiber optic installations for optimizing production and maximizing reservoir recovery.

Nomenclature

Parameter	Definition
A	Cross-sectional area open to flow, m^2
c_i	Zonal contribution coefficient for zone i
C	Correlation coefficient, dimensionless
M	Total number of time steps in the dataset
n	Power-law exponent, dimensionless
N	Peak-to-peak noise level, mV
P_i	Acoustic energy for zone i at time t , linear units (e.g., RMS strain)
q_i	Zonal volumetric flow rate for zone i at time t , m^3/s
Q	Total bottom-hole volumetric flow rate at time t , m^3/s
ρ	Fluid density, kg/m^3

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